

INVERSE TRIGONOMETRY – THEORY & FORMULAS

1. What are inverse trigonometric functions?

- The usual trigonometric functions $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$ are many-one on $\mathbb{R}$, so they do not have inverses on the whole real line.^{[1][2]}
- By restricting their domains to suitable intervals where they become one-one and onto, we can define their inverses, called inverse trigonometric functions.^{[3][1]}

For each basic trig function f , the inverse function f^{-1} or $\arcsin$, $\arccos$, $\arctan$, etc. returns an angle (in radians) whose trig value is the given number.

Standard notation (Class 12 / JEE pattern):

- $\sin^{-1} x$ or $\arcsin x$
- $\cos^{-1} x$ or $\arccos x$
- $\tan^{-1} x$ or $\arctan x$
- $\cot^{-1} x, \sec^{-1} x, \csc^{-1} x$ ^{[2][3]}

2. Principal values, domains and ranges

Each inverse trig function is defined with a **principal value branch**, i.e., restricted range so that it becomes a single-valued function.^{[4][1][3]}

2.1 Standard principal branches

1. Inverse sine

- Definition: $y = \sin^{-1} x \Leftrightarrow x = \sin y, y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.^{[5][1]}
- Domain: $x \in [-1, 1]$ ^{[3][4]}
- Range: $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

2. Inverse cosine

- Definition: $y = \cos^{-1} x \Leftrightarrow x = \cos y, y \in [0, \pi]$.^{[1][5]}
- Domain: $x \in [-1, 1]$.
- Range: $y \in [0, \pi]$.^{[4][3]}

3. Inverse tangent

- Definition: $y = \tan^{-1} x \Leftrightarrow x = \tan y, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.^{[1][3]}
- Domain: all real $x \in \mathbb{R}$.
- Range: $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.^{[3][4]}

4. Inverse cotangent (common convention in JEE/NCERT)

- Definition: $y = \cot^{-1} x \Leftrightarrow x = \cot y, y \in (0, \pi)$.^{[2][5]}
- Domain: all real x .
- Range: $y \in (0, \pi)$.

5. Inverse secant

- Definition: $y = \sec^{-1} x \Leftrightarrow x = \sec y, y \in [0, \pi], y \neq \frac{\pi}{2}$.^{[2][3]}
- Domain: $x \in (-\infty, -1] \cup [1, \infty)$.
- Range: $y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$.

6. Inverse cosecant

- Definition: $y = \csc^{-1} x \Leftrightarrow x = \csc y, y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$.^{[3][2]}
- Domain: $x \in (-\infty, -1] \cup [1, \infty)$.
- Range: $y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$.

3. Basic values (standard angles)

Using the definitions and principal ranges, we get standard exact values.^{[6][2]}

3.1 $\sin^{-1} x$

- $\sin^{-1} 0 = 0$

- $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$
- $\sin^{-1} \frac{\sqrt{2}}{2} = \frac{\pi}{4}$
- $\sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$
- $\sin^{-1} (-a) = -\sin^{-1} a$ for $a \in [-1,1]$.

3.2 $\cos^{-1} x$

- $\cos^{-1} 1 = 0$
- $\cos^{-1} \frac{1}{2} = \frac{\pi}{3}$
- $\cos^{-1} 0 = \frac{\pi}{2}$
- $\cos^{-1} \left(-\frac{1}{2}\right) = \frac{2\pi}{3}$.

3.3 $\tan^{-1} x$

- $\tan^{-1} 0 = 0$
- $\tan^{-1} 1 = \frac{\pi}{4}$
- $\tan^{-1} \sqrt{3} = \frac{\pi}{3}$
- $\tan^{-1} (-x) = -\tan^{-1} x$.

Similar values exist for $\cot^{-1} x$, $\sec^{-1} x$, $\csc^{-1} x$ using principal ranges.^{[5][2]}

4. Symmetry properties (odd/even type)

These follow from the symmetry of the original trig functions and the principal ranges.^{[7][8][9]}

1. Inverse sine

$$\sin^{-1} (-x) = -\sin^{-1} x \text{ (odd).}$$

2. Inverse tangent

$$\tan^{-1} (-x) = -\tan^{-1} x \text{ (odd).}$$

3. Inverse cosine

$$\cos^{-1} (-x) = \pi - \cos^{-1} x \text{ (neither odd nor even).}$$

4. **Inverse cotangent** (principal $0 < y < \pi$)

$$\cot^{-1}(-x) = \pi - \cot^{-1} x.$$

5. **Inverse secant**

$$\sec^{-1}(-x) = \pi - \sec^{-1} x \text{ for } |x| \geq 1.$$

6. **Inverse cosecant**

$$\csc^{-1}(-x) = -\csc^{-1} x \text{ (odd-type on the chosen branch).}$$

These are widely used in simplification and solving equations.^{[10][7]}

5. Fundamental relationships between inverse functions

5.1 Complementary-type identities

- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for all $x \in [-1, 1]$.^{[7][10]}
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ for all $x > 0$ (with the standard branches).^{[10][7]}
- $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$ for $x > 1$.^[2]

These are obtained by converting both sides to an angle θ and using right-triangle geometry.

5.2 Conversion to $\tan^{-1}$ form

For $|x| < 1$:^{[7][9]}

- $\sin^{-1} x = \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right).$
- $\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$ for $0 < x \leq 1$.

These come from taking $x = \sin \theta$ or $x = \cos \theta$ and using $\tan \theta$.

6. Sum and difference formulas

The most used ones are for $\tan^{-1}$.^{[11][7]}

6.1 For $\tan^{-1}$

For real x, y with suitable conditions:

- $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ if $xy < 1$ and the resulting angle lies in the principal interval.^{[9][7]}
- $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$ if $xy > -1$ and the angle is in the principal interval.

These identities are central for JEE-type telescoping sums and tricky evaluations.^{[12][11]}

6.2 For $\sin^{-1}$

When values lie in valid ranges:^{[9][7]}

- $\sin^{-1} x + \sin^{-1} y = \sin^{-1} (x\sqrt{1-y^2} + y\sqrt{1-x^2})$ under conditions such as $x^2 + y^2 \leq 1$ and $xy \geq 0$.
- $\sin^{-1} x - \sin^{-1} y = \sin^{-1} (x\sqrt{1-y^2} - y\sqrt{1-x^2})$ with similar restrictions.

6.3 For $\cos^{-1}$

Analogous formulas:^{[9][7]}

- $\cos^{-1} x + \cos^{-1} y = \cos^{-1} (xy - \sqrt{(1-x^2)(1-y^2)})$ for appropriate x, y .
- $\cos^{-1} x - \cos^{-1} y = \cos^{-1} (xy + \sqrt{(1-x^2)(1-y^2)})$.

These are used less often at board level but appear in advanced problems.^{[11][7]}

7. Composition with direct trigonometric functions

Very common pattern: $\sin(\sin^{-1} x)$, $\sin^{-1}(\sin x)$, etc.^{[8][3]}

7.1 “Inside then outside” (easy direction)

For all x in the domain of the inverse function:

- $\sin(\sin^{-1} x) = x, x \in [-1, 1]$.
- $\cos(\cos^{-1} x) = x, x \in [-1, 1]$.
- $\tan(\tan^{-1} x) = x, x \in \mathbb{R}$.^{[11][3]}

Similarly for $\cot, \sec, \csc$ on their domains.

7.2 “Outside then inside” (careful with range)

Because of principal branches:

- $\sin^{-1}(\sin x)$ is not always x ; it equals the unique angle $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ with $\sin y = \sin x$.^{[13][11]}
- $\cos^{-1}(\cos x)$ is the unique $y \in [0, \pi]$ with $\cos y = \cos x$.^{[13][11]}
- $\tan^{-1}(\tan x)$ is the unique $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with $\tan y = \tan x$.^{[13][11]}

These are key for principal value evaluation questions.

8. Useful algebraic expressions in terms of x

Assume $|x| \leq 1$ where needed.^{[2][9]}

1. From right-triangle definitions:

- $\sin(\cos^{-1} x) = \sqrt{1-x^2}$ (taking positive root in the principal range).
- $\cos(\sin^{-1} x) = \sqrt{1-x^2}$.
- $\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}$.
- $\sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}$.
- $\cos(\tan^{-1} x) = \frac{1}{\sqrt{1+x^2}}$.^{[2][3]}

2. Similarly:

- $\cot(\sin^{-1} x) = \frac{\sqrt{1-x^2}}{x}$ (for $x \neq 0$).
- $\tan(\cos^{-1} x) = \frac{\sqrt{1-x^2}}{x}$ (for $x > 0$).
- $\sec(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$.
- $\csc(\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}}$.^{[9][2]}

These forms are heavily used in calculus questions (limits, derivatives, integrals).

9. Important derived identities (JEE focus)

Some standard target forms:^{[12][7]}

1. Double-angle linkage:

- If $\theta = \sin^{-1} x$, then $\cos(2\theta) = 1 - 2x^2$.
- If $\theta = \tan^{-1} x$, then $\sin(2\theta) = \frac{2x}{1+x^2}$ and $\cos(2\theta) = \frac{1-x^2}{1+x^2}$.

2. Parametric substitutions:

- Put $x = \sin \theta$, so $\theta = \sin^{-1} x$.
- Put $x = \cos \theta$, so $\theta = \cos^{-1} x$.
- Put $x = \tan \theta$, so $\theta = \tan^{-1} x$.

These substitutions reduce many inverse trig problems to simple trig.^{[5][1]}

3. Relations between inverse expressions:

- $\sin^{-1} \left(\frac{2x}{1+x^2} \right) = 2\tan^{-1} x$ on suitable intervals.
- $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = 2\tan^{-1} x$ for $x > 0$.
- $\tan^{-1} \left(\frac{2x}{1-x^2} \right) = 2\tan^{-1} x$ where defined.^{[11][7]}

These come from half-angle and tangent-of-sum formulas.

10. Graphs (qualitative)

Inverse trig graphs are reflections of the basic trig graphs (restricted) in the line $y = x$.^{[13][3]}

1. $y = \sin^{-1} x$

- Domain: $[-1, 1]$.
- Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
- Increasing, passes through origin, S-shaped curve.^{[4][3]}

2. $y = \cos^{-1} x$

- Domain: $[-1, 1]$.
- Range: $[0, \pi]$.

- Decreasing, passes through $(1,0)$ and $(-1,\pi)$.^{[4][3]}

3. $y = \tan^{-1} x$

- Domain: $\mathbb{R}$.
- Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.
- Increasing; horizontal asymptotes $y = \pm \frac{\pi}{2}$.^{[3][13]}

4. **Others** ($\cot^{-1}, \sec^{-1}, \csc^{-1}$)

- Defined similarly with appropriate discontinuities and asymptotes based on their domains.^{[4][3]}

Graphs are important for understanding principal values and solving inequality-type questions.

11. Calculus links (brief)

Inverse trig functions are frequently used in derivatives and integrals.^{[8][6]}

11.1 Standard derivatives (for $|x| < 1$ where required)

- $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$.
- $\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$.
- $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$.
- $\frac{d}{dx} \cot^{-1} x = -\frac{1}{1+x^2}$.
- $\frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$ for $|x| > 1$.
- $\frac{d}{dx} \csc^{-1} x = -\frac{1}{|x|\sqrt{x^2-1}}$ for $|x| > 1$.^{[8][9]}

11.2 Standard integrals

- $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$.
- $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$.
- Variants with algebraic substitutions reduce to these two master forms.^{[8][9]}

12. Typical problem types (NCERT + JEE)

Common question styles built from this theory:^{[14][12][5]}

1. **Evaluate principal values** of expressions like $\sin^{-1}(\sin 5)$, etc.
2. **Simplify algebraic combinations** using identities: $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$, telescoping sums of $\tan^{-1}$.
3. **Solve equations/inequalities** involving inverse trig, e.g. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, or $\sin^{-1} x \leq \frac{\pi}{4}$.
4. **Express algebraic functions as trig/inverse trig** using substitutions $x = \sin \theta, \cos \theta, \tan \theta$.
5. **Use inverse trig in calculus**: limits like $\frac{\sin^{-1} x}{x}$ as $x \rightarrow 0$, derivatives of compositions, and standard integrals.

These patterns are exactly what your LEVEL1–4 question files are designed to practice.^{[15][6][10]}

If you want, the theory can also be arranged in NCERT-style subsections (2.1, 2.2, etc.) with small worked examples next.

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1. https://en.wikipedia.org/wiki/Inverse_trigonometric_functions
2. <https://byjus.com/maths/inverse-trigonometric-functions/>
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